BERNSTEIN
SOCIETE GENERALE GROUP

15 October 2026

U.S. Multi Industry & Electrical Equipment

Data Center Equipment: Sizing the US Cooling TAM

Varun Govindaraj
+1 917 344 8543
varun.govindaraj@bernsteinsg.com

Chad Dillard
+1 917 344 8469
chad.dillard@bernsteinsg.com

Madison Rezaei
+1 917 344 8622
madison.rezaei@bernsteinsg.com

Alasdair Leslie
+44 20 7762 4952
alasdair.leslie@bernsteinsg.com

Alex Wang, CFA
+852 2123 2613
alex.wang@bernsteinsg.com

Specialist Sales

Steve Song
+1 917 344 8401
steve.song@bernsteinsg.com

The data center cooling equipment market is relatively opaque, with a fragmented set of sizing estimates. In this note, we attempt to size different types of equipment to better reflect an ever-changing growth outlook.

Methodology: We start with the Bernstein US data center forecast and use incremental facility GW additions as the foundation for our TAM. We then forecast the mix of liquid-cooled vs. air-cooled deployments for these additions and apply our updated cooling content assumptions ($M/MW). This gives us the TAM for cooling components. As a final step, we sanity check our numbers against public disclosures / bottom-up market sizing across the largest players in each equipment category. Model attached here.

Key takeaways: We estimate the overall US data center cooling TAM at **~$25B in 2026E**, growing to **~$63B by 2030E (~25% 4YCAGR)**. Chillers remain the largest category at **~$10B in 2026E**, reaching **~$23B by 2030E** (~25%+ 4YCAGR). We estimate the CDU market at **~$5B**, growing to **~$14B** (~30%+ 4YCAGR); *notably, we have increased our prior CDU estimates,* reflecting updated assumptions *and our view that Delta has emerged as a key supplier to the US data center market through indirect shipments of liquid-to-air (L2A) CDUs to cloud service providers (CSPs), which is now reflected in our sizing.* Cold plates grow from **~$3B to ~$10B** (~30%+ 4YCAGR), while manifolds increase from **~$1B to ~$3B** (~30%+ 4YCAGR). By comparison, CRAHs and RDHx grow more slowly, from **~$2B to ~$3B** each (~15% 4Y CAGR) as the share of air cooling declines across incremental deployments. **Overall, we expect liquid cooling equipment (CDUs, cold plates, manifolds) and chillers to outgrow incremental GW additions, driven by rising liquid cooling penetration and pricing increases amid tight supply chains and rapidly growing demand.**

From a pure liquid-cooling torque perspective, we believe that **Vertiv, nVent and Delta** offer the most direct exposure to benefit from the transition from air cooling to liquid cooling in our coverage universe. This is not to say that other companies highlighted in the report do not benefit (particularly as modular / prefabricated deployments gain traction). We would also like to note that Asia Vital Components (3017.TW; not covered) appears well positioned from a liquid cooling exposure standpoint.

Where could we be wrong? Recent industry developments (such as ~45°C+ inlet temperatures), suggest a future architecture where the reliance on chillers *could be* reduced, an area we continue to monitor closely. Our TAM estimates are also higher than previously published third-party forecasts; however, this reflects both our more refined view on top-down data center capacity additions and the fact that our estimates are triangulated against bottom-up market sizing for major component suppliers.

While we acknowledge that the data center cooling ecosystem remains opaque and precise estimation remains challenging, the combination of top-down and bottom-up approaches gives us confidence in the broader direction of our TAM forecast. We welcome engagement with investors who hold alternative views and expect our estimates to continue evolving as both industry disclosures and our own understanding are refined over time.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our Global Research Disclosure Website.

First Published: 15 Oct 2026 04:01 UTC Completion Date: 14 Oct 2026 22:26 UTC

BERNSTEIN TICKER TABLE

			14 Oct 2026		TTM	Adjusted EPS				Adjusted P/E (x)		
Ticker	Rating	Cur	Closing Price	Price Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
VRT (Vertiv)	O	USD	222.83	368.00	11.0%	USD	4.20	6.60	9.06	53.1	33.8	24.6
NVT (nVent)	O	USD	152.72	229.00	37.3%	USD	3.34	5.15	6.55	45.7	29.7	23.3
TT (Trane)	O	USD	471.29	575.00	(5.0)%	USD	13.06	15.28	17.79	36.1	30.8	26.5
JCI (Johnson Controls)	O	USD	152.44	190.00	25.2%	USD	3.78	5.09	6.35	40.3	30.0	24.0
CARR (Carrier)	M	USD	54.24	64.00	(21.8)%	USD	2.57	2.86	3.24	21.1	18.9	16.8
2308.TT (Delta)	O	TWD	1,785.00	2,620.00	58.7%	TWD	23.09	39.35	57.98	77.3	45.4	30.8
ETN (Eaton)	O	USD	408.85	534.00	(7.0)%	USD	12.07	13.45	16.37	33.9	30.4	25.0
DLR (Digital Realty)	O	USD	172.09	226.00	(16.1)%	USD	3.87	3.09	3.21	44.5	55.6	53.7
EQIX (Equinix)	O	USD	995.90	1,270.00	5.7%	USD	14.96	17.12	20.13	66.6	58.2	49.5
BXDC (Blackstone Digital Infra)	O	USD	16.28	26.00	NA	USD	0.00	0.32	(1.00)		51.0	(16.2)
CSQR (Csquare)	O	USD	16.82	27.00	NA	USD	0.00	(1.34)	(1.06)		(12.5)	(15.8)
SU.FP (Schneider)	O	EUR	264.40	350.00	(4.3)%	EUR	8.43	10.42	13.24	31.4	25.4	20.0
SPX			7,718.94									
ASIAX			1,876.06									
EDME			1,581.51									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

2308.TT estimate is Reported EPS; 2308.TT valuation is Reported P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate VRT (PT $368), NVT (PT $229), Trane (PT $575) and JCI (PT $190) Outperform.

We rate CARR (PT $64) Market-Perform.

We rate Delta Electronics Outperform, with PT = NT$2,620.

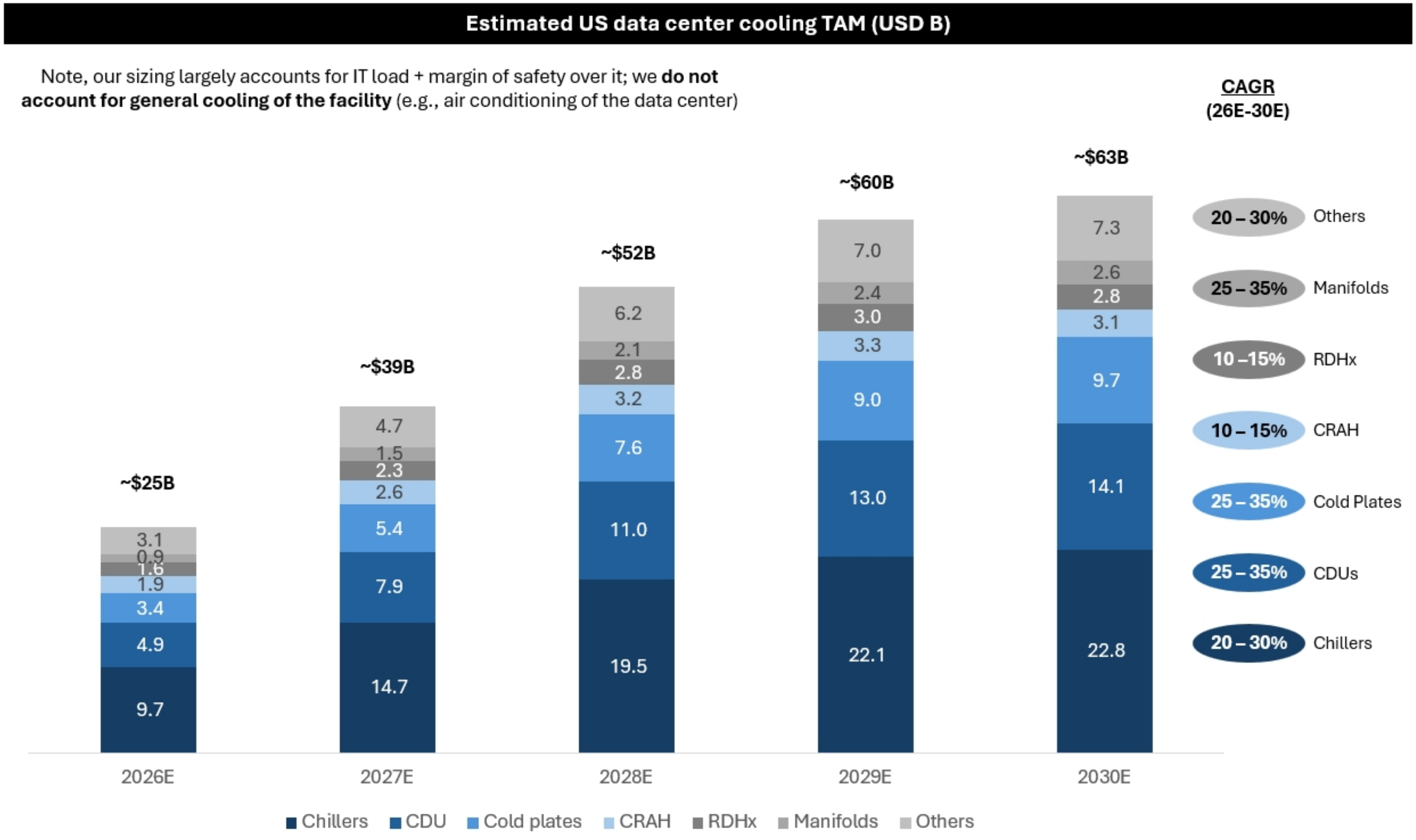
We rate Schneider Outperform, with a price target of €350.

We rate ETN Outperform, with PT $534.00.

We rate DLR, EQIX, BXDC, and CSQR Outperform.

DETAILS

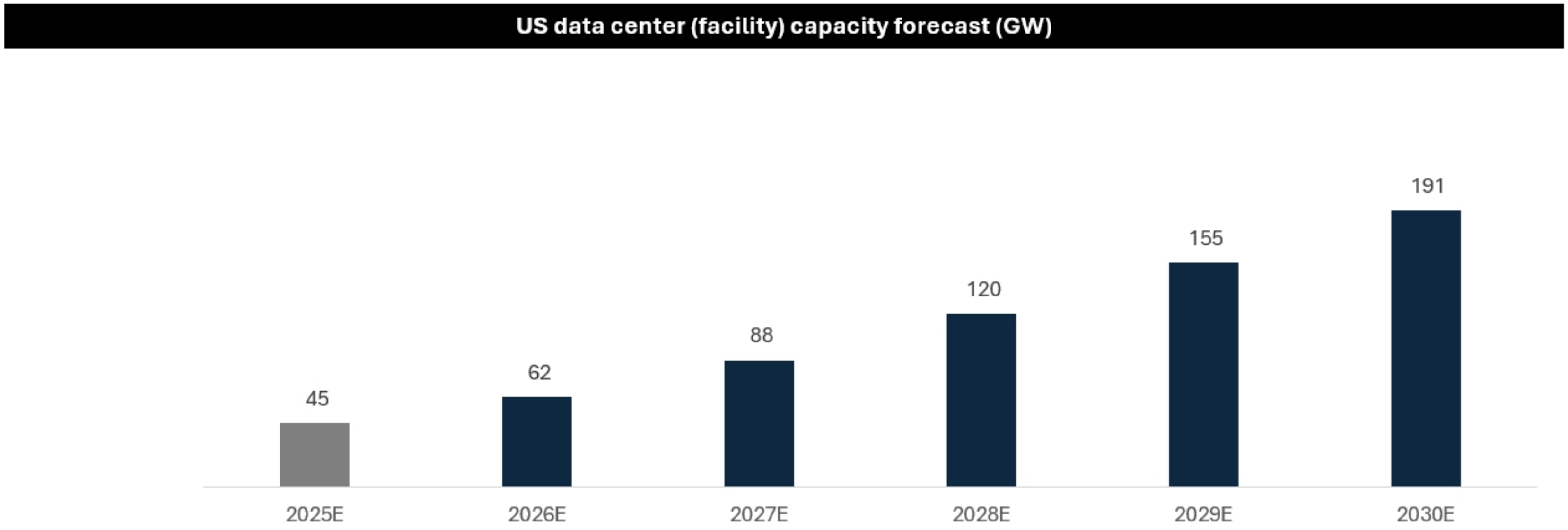
EXHIBIT 1: **We estimate total US data center cooling TAM at ~$25B in 2026E to reach ~$63B by 2030E at ~25% 4Y CAGR**



Note, our sizing largely accounts for IT load + margin of safety over it; we **do not account for general cooling of the facility** (e.g., air conditioning of the data center)

Source: Bernstein Analysis and Estimates

EXHIBIT 2: **US data center capacity (facility GW) forecasted to grow to ~190GW by 2030E**

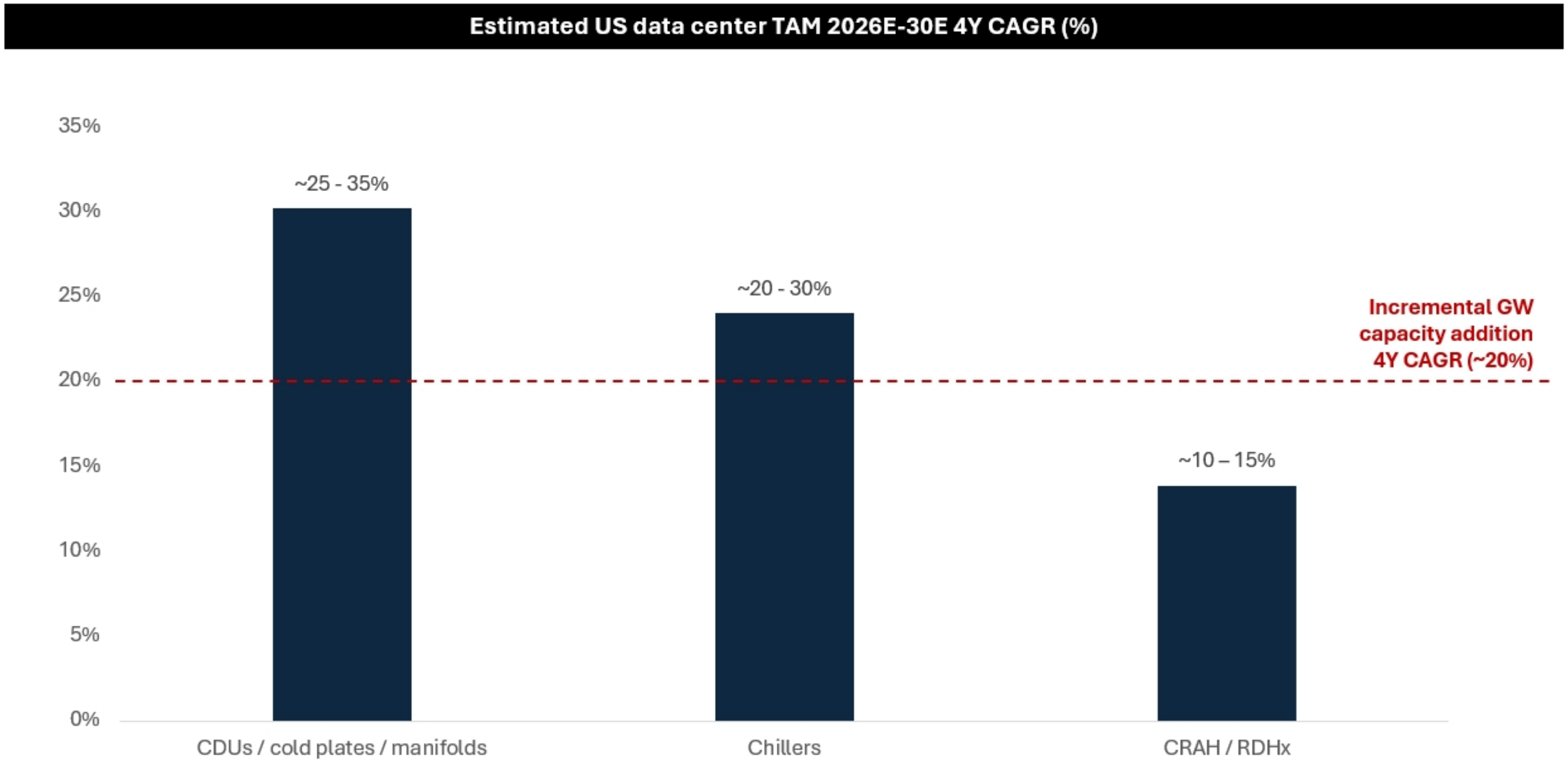


	2025E	2026E	2027E	2028E	2029E	2030E
Incremental capacity added *(GW)*	-	~18	~25	~32	~35	~35
% of liquid cooled *(incremental capacity)*	-	~70%	~74%	~78%	~81%	~85%
% of air cooled *(incremental capacity)*	-	~30%	~26%	~22%	~19%	~15%

Source: Bernstein Data Center Forecast Model, Bernstein Analysis and Estimates

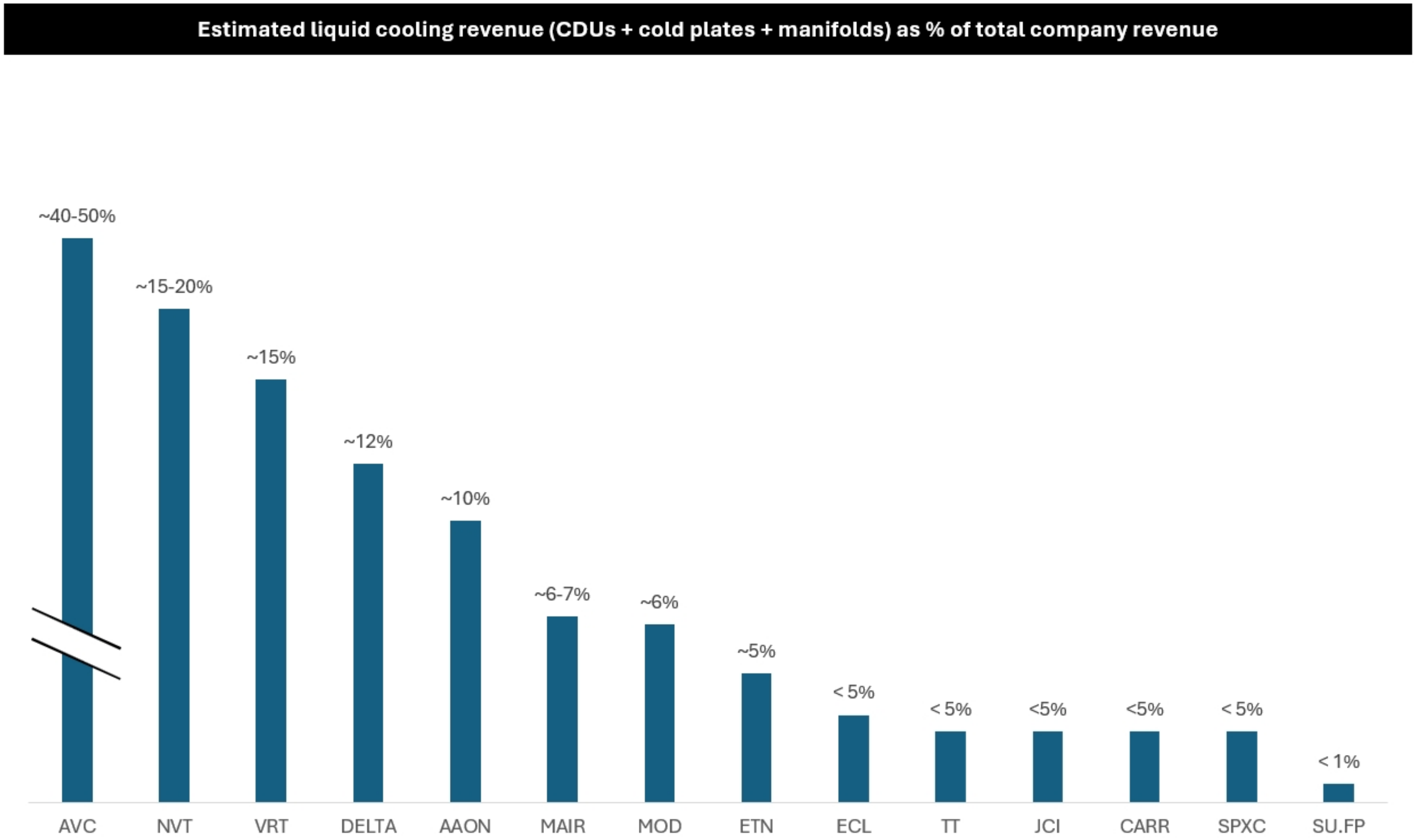
Varun Govindaraj +1 917 344 8543 varun.govindaraj@bernsteinsg.com

EXHIBIT 3: **Liquid cooling equipment (CDUs, cold plates, manifolds) and chillers expected to grow faster than GW additions, driven by rising liquid cooling penetration and price increases**



Source: Bernstein Analysis and Estimates

EXHIBIT 4: **Some companies are more exposed to liquid cooling than others**



Source: Bernstein Analysis and Estimates

EXHIBIT 5: **As an equipment provider to data centers, how can you get "torque" over incremental GW? Three themes stand out.**

Liquid cooling

Driven by shift of cooling mechanism in the white space of a data center from air to liquid using equipment like CDUs + cold plates vs. CRAHs / RDHx for a bulk of heat transfer

Focus of this note

Modular / pre-fab

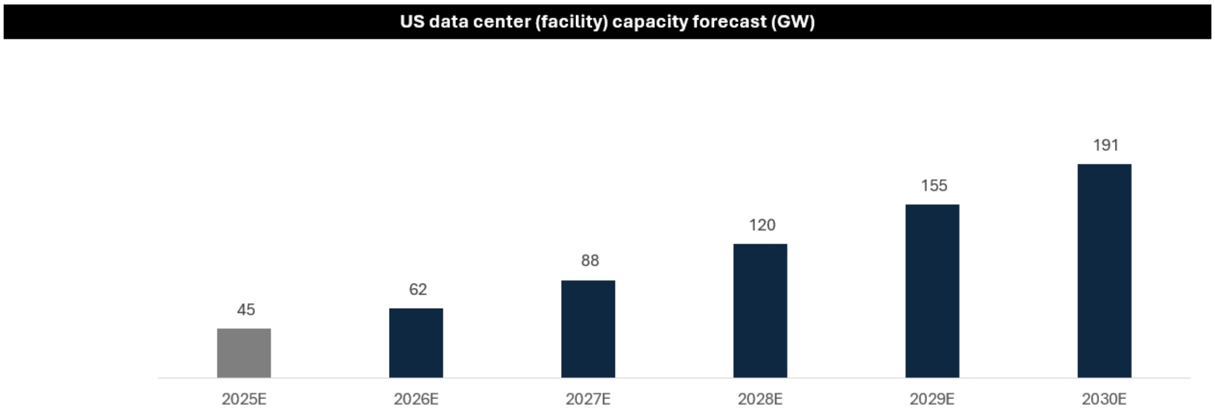
Driven by shift from "stick built" (which implies point solutions) to "modular" solutions where multiple products are packaged to develop a holistic offering that can be set up and installed faster

800V DC

Forward looking (2028+), but a shift in the underlying power architecture in the DC that moves away from AC to DC input; ideally leads to simplification / elimination of components (e.g., UPS)

Source: Bernstein Analysis

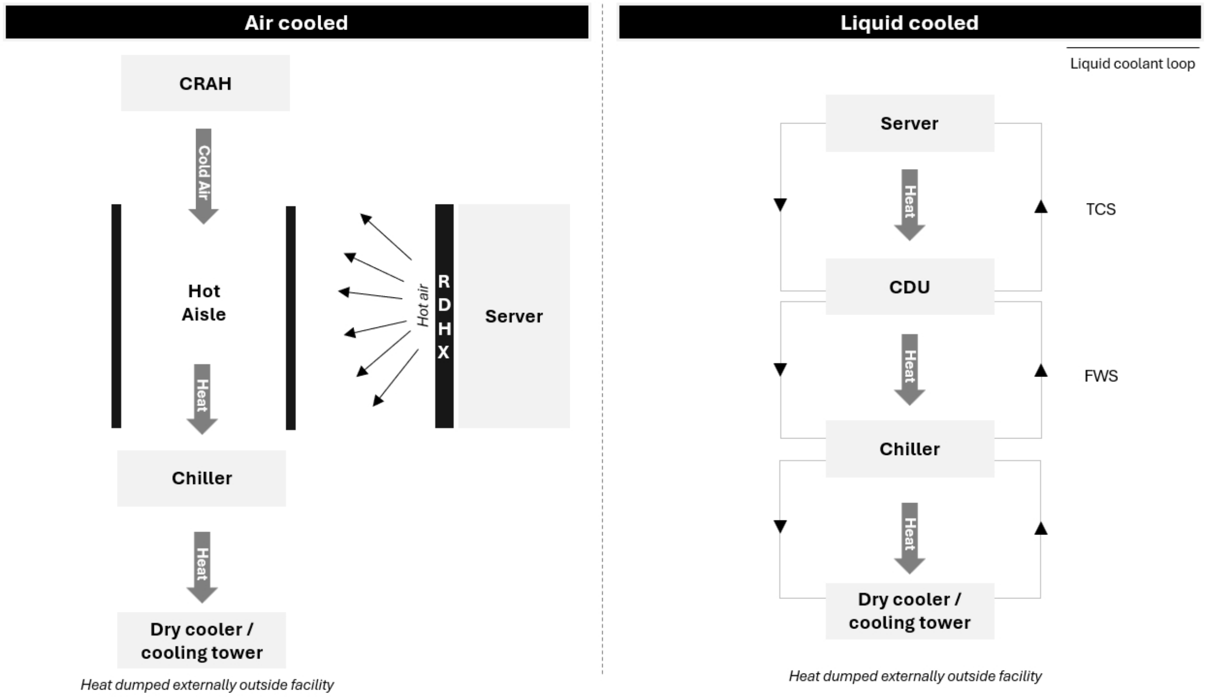
EXHIBIT 6: **US data center capacity (facility GW) forecasted to grow to ~190GW by 2030E**



	2025E	2026E	2027E	2028E	2029E	2030E
Incremental capacity added *(GW)*	-	~18	~25	~32	~35	~35
% of liquid cooled *(incremental capacity)*	-	~70%	~74%	~78%	~81%	~85%
% of air cooled *(incremental capacity)*	-	~30%	~26%	~22%	~19%	~15%

Source: Bernstein Data Center Forecast model, Bernstein Analysis and Estimates

EXHIBIT 7: **Recap | Different cooling architectures for data centers**


Air cooled
CRAH
Cold Air
Hot Aisle
Heat
Hot air
RDHX
Server
Chiller
Heat
Dry cooler / cooling tower
Heat dumped externally outside facility
Liquid cooled
Liquid coolant loop
Server
Heat
TCS
CDU
Heat
FWS
Chiller
Heat
Dry cooler / cooling tower
Heat dumped externally outside facility

Source: Bernstein Analysis

EXHIBIT 8: **Components needed for cooling in data centers (1/2)**

✔ Required ✓ Required at partial capacity ✗ Not required

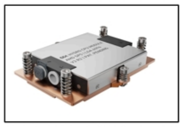
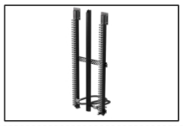
Component	Description		Value ($M/MW)	Requirement in architecture	
				Air-cooled	Liquid-cooled
Chiller[1]		**"Central engine" of the cooling plant,** responsible for cooling water to inject into TCS and rejecting heat from warm liquid to the FWS	~0.50 – 0.60	✔	✔
Dry cooler		**"Heat fans"** to blow hot air outside of the facility	~0.10 – 0.20	✔	✔
Cooling tower		**"Heat rejection equipment"** sitting outside the facility to transfer heat from facility water system to environment	~0.10 – 0.20	✔	✔
Computer room air handler (CRAH)[2]		**Data-hall-level air handler** that cools residual hot air over chilled-water coils	~0.40 – 0.60	✔	✓

1. Fully loaded cost including chiller, installation labor, pipes, pumping, heat rejection equipment (dry coolers and cooling towers) 2. While gross unit price is shown, we have taken residual heat assumptions while calculating overall content cost for both air-cooled and liquid-cooled

Source: Bernstein Analysis and Estimates

EXHIBIT 9: **Components needed for cooling in data centers (2/2)**

✔ Required ✓ Required at partial capacity ✗ Not required

Component		Description	Value ($M/MW)	Requirement in architecture	
				Air-cooled	Liquid-cooled
Rear door heat exchanger (RDHx)[1]		**Server rack-level liquid-to-air coil** that rejects heat exhaust from the back of the rack	~0.30 – 0.50	✔	✓
Coolant distribution unit (CDU)[2]		**“Gateway” between facility water and chip-level liquid cooling loop;** controls coolant flow, pressure and temperature to liquid-cooled racks	~0.35 – 0.45	✗	✔
Cold plates		**“Metal plates” mounted on GPUs / CPUs,** responsible for pulling heat directly from chip into the liquid loop	~0.25 – 0.30	✗	✔
Manifolds		**Rack plumbing** that distributes coolant from CDU to each cold plate and returns the warm liquid	~0.05 – 0.10	✗	✔
Data center infrastructure management (DCIM) software		**Monitoring and control software** that tracks the entire ecosystem	~0.05 – 0.10	✔	✔

1. While gross unit price is shown, we have taken residual heat assumptions while calculating overall content cost for both air-cooled and liquid-cooled 2. All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation

Source: Bernstein Analysis and Estimates

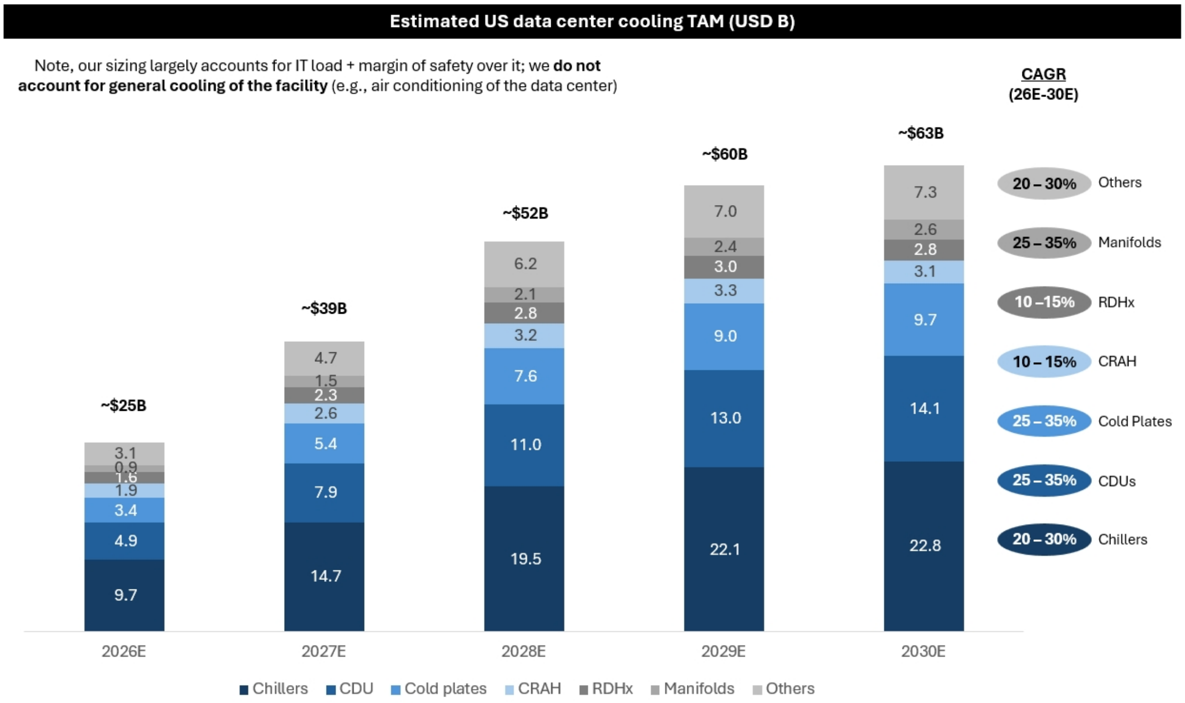
EXHIBIT 10: **Total cooling content opportunity by architecture ($M / MW)**

Component		Air cooled	Liquid cooled	Remarks
Cooling	**Chillers**[1]	~0.50 – 0.60	~0.50 – 0.60	*Revised by ~10% upwards as prior estimate was conservative*
	CRAH[2]	~0.20 – 0.30	~0.04 – 0.05	*Limited to compute cooling (GPU heat load); excludes facility cooling*
	RDHx[2]	~0.15 – 0.25	~0.04 – 0.05	*As per prior estimate (Cooling Content $M / MW note)*
	CDU[3]	NA	~0.35 – 0.45	*Revised by ~10% upwards as prior estimate was conservative*
	Cold plates	NA	~0.25 – 0.30	*Revised by ~10% upwards as prior estimate was conservative*
	Manifolds	NA	~0.05 – 0.10	*As per prior estimate (Cooling Content $M / MW note)*
	DCIM software	~0.05 – 0.10	~0.05 – 0.10	*As per prior estimate (Cooling Content $M / MW note)*
Ancillaries (incl. refrigerant)		~0.10	~0.10	*As per prior estimate (Cooling Content $M / MW note)*
Total Opportunity ($M / MW)		**~1.0 – 1.4**	**~1.3 – 1.7**	*Excludes redundancy assumptions; baked into facility GW forecast*

1. Fully loaded cost including chiller, installation labor, pipes, pumping, heat rejection equipment (dry coolers and cooling towers) 2. For air-cooled: assuming required at 50% of gross unit value as residual heat consumption is split 50/50 among CRAH and RDHx; For liquid-cooled: assuming ancillary heat capture split 50/50 among CRAH and RDHx 3. All-in costs for pure-play liquid-to-liquid CDUs including filtration, integration, labor and installation

Source: Bernstein Analysis and Estimates

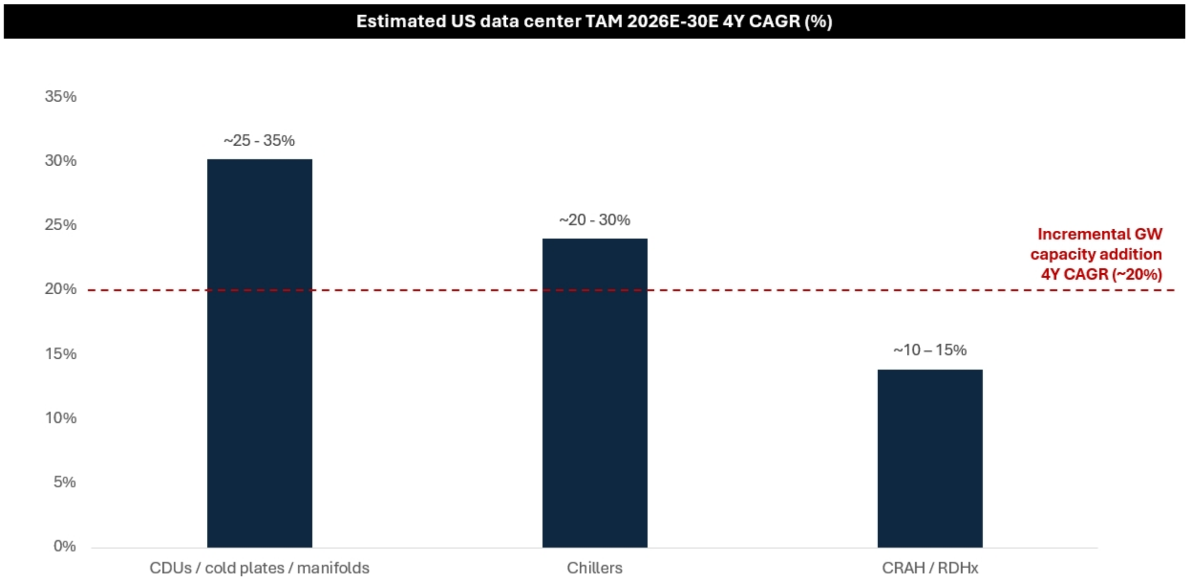
EXHIBIT 11: **We estimate total US data center cooling TAM at ~$25B in 2026E to reach ~$63B by 2030E at ~25% 4Y CAGR**


Estimated US data center cooling TAM (USD B)

Note, our sizing largely accounts for IT load + margin of safety over it; we **do not account for general cooling of the facility** (e.g., air conditioning of the data center)

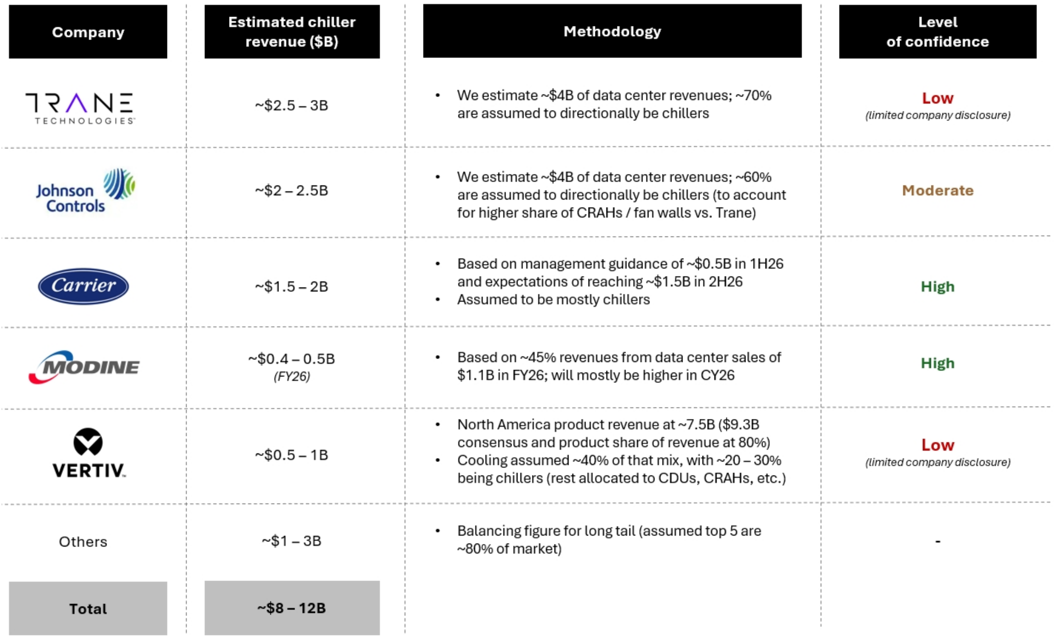
Source: Bernstein Analysis and Estimates

EXHIBIT 12: **Liquid cooling equipment (CDUs, cold plates, manifolds) and chillers expected to grow faster than GW additions, driven by rising liquid cooling penetration and price increases**


Estimated US data center TAM 2026E-30E 4Y CAGR (%)

Source: Bernstein Analysis and Estimates

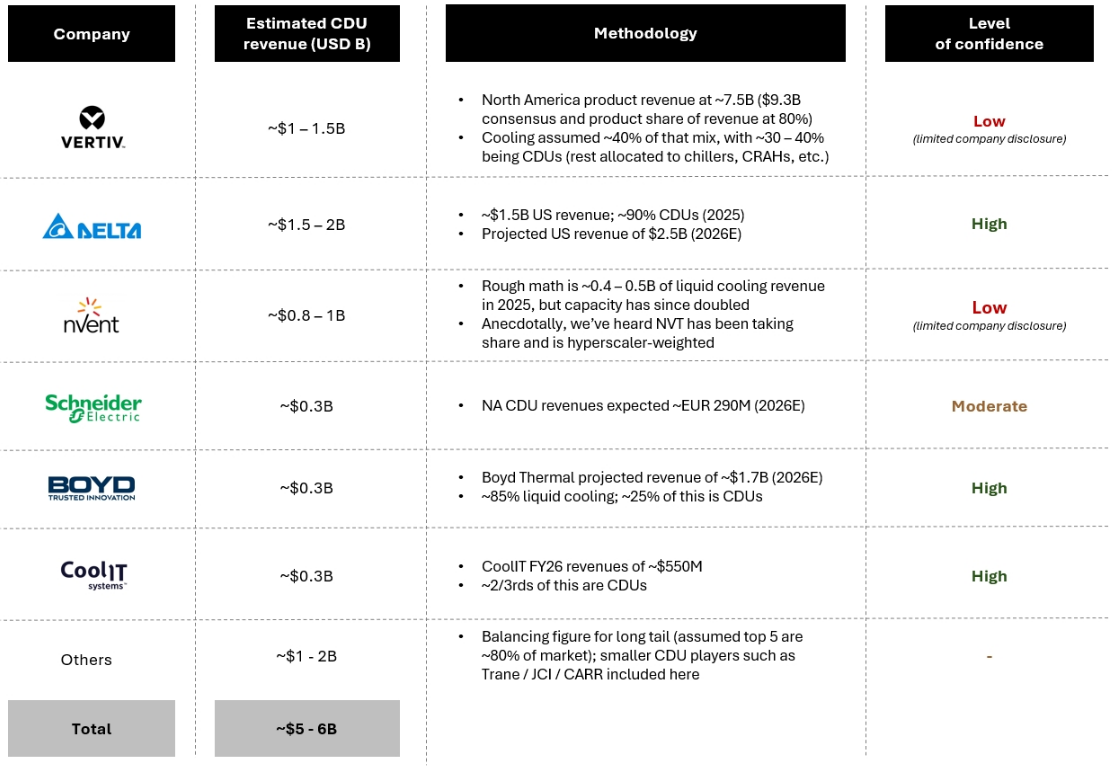
EXHIBIT 13: **Sanity check of chiller market size (North America only)**



Company	Estimated chiller revenue ($B)	Methodology	Level of confidence
TRANE TECHNOLOGIES	~$2.5 – 3B	We estimate ~$4B of data center revenues; ~70% are assumed to directionally be chillers	Low *(limited company disclosure)*
Johnson Controls	~$2 – 2.5B	We estimate ~$4B of data center revenues; ~60% are assumed to directionally be chillers (to account for higher share of CRAHs / fan walls vs. Trane)	Moderate
Carrier	~$1.5 – 2B	Based on management guidance of ~$0.5B in 1H26 and expectations of reaching ~$1.5B in 2H26 Assumed to be mostly chillers	High
MODINE	~$0.4 – 0.5B *(FY26)*	Based on ~45% revenues from data center sales of $1.1B in FY26; will mostly be higher in CY26	High
VERTIV	~$0.5 – 1B	North America product revenue at ~7.5B ($9.3B consensus and product share of revenue at 80%) Cooling assumed ~40% of that mix, with ~20 – 30% being chillers (rest allocated to CDUs, CRAHs, etc.)	Low *(limited company disclosure)*
Others	~$1 – 3B	Balancing figure for long tail (assumed top 5 are ~80% of market)	-
Total	**~$8 – 12B**		

Source: Bernstein Analysis and Estimates

EXHIBIT 14: **Sanity check of CDU market size (North America only)**



Company	Estimated CDU revenue (USD B)	Methodology	Level of confidence
VERTIV	~$1 – 1.5B	North America product revenue at ~7.5B ($9.3B consensus and product share of revenue at 80%) Cooling assumed ~40% of that mix, with ~30 – 40% being CDUs (rest allocated to chillers, CRAHs, etc.)	Low *(limited company disclosure)*
DELTA	~$1.5 – 2B	~$1.5B US revenue; ~90% CDUs (2025) Projected US revenue of $2.5B (2026E)	High
nVent	~$0.8 – 1B	Rough math is ~0.4 – 0.5B of liquid cooling revenue in 2025, but capacity has since doubled Anecdotally, we've heard NVT has been taking share and is hyperscaler-weighted	Low *(limited company disclosure)*
Schneider Electric	~$0.3B	NA CDU revenues expected ~EUR 290M (2026E)	Moderate
BOYD TRUSTED INNOVATION	~$0.3B	Boyd Thermal projected revenue of ~$1.7B (2026E) ~85% liquid cooling; ~25% of this is CDUs	High
CoolIT systems	~$0.3B	CoolIT FY26 revenues of ~$550M ~2/3rds of this are CDUs	High
Others	~$1 - 2B	Balancing figure for long tail (assumed top 5 are ~80% of market); smaller CDU players such as Trane / JCI / CARR included here	-
Total	**~$5 - 6B**		

Source: Bernstein Analysis and Estimates

EXHIBIT 15: **Sanity check of cold plates market size (North America only)**

Company	Estimated cold plates revenue ($B)	Methodology	Level of confidence
BOYD TRUSTED INNOVATION	~$0.9 – 1B	• Boyd Thermal projected revenue of ~$1.7B (2026E) • ~85% liquid cooling; ~65% of this is cold plates	High
CoolIT systems	~$0.2 – 0.3B	• CoolIT FY26 revenues of ~$550M • ~1/3rds of this is cold plates	High
AVC	~$1 – 1.5B	• AVC FY26E revenue ~$7B; ~45% liquid cooling • ~70% North America revenue • 70/30 split among cold plates & manifolds	Moderate
Auras	~$0.2 – 0.3B	• Auras FY26E revenue ~$1.2B; ~50% liquid cooling • ~60% North America revenue • 80/20 split among cold plates & manifolds	Moderate
COOLER MASTER	~$0.5 – 1B	• Assuming slightly lesser market share to AVC	Low *(limited company disclosure)*
DELTA	~$0.1 – 0.3B	• Projected US revenue of $2.5B (2026E) • ~10% cold plates revenue	Moderate
Others	~$0.3 – 0.5B	• Balancing figure for long tail (assumed top 5 are ~80% of market)	-
Total	**~$3 – 5B**		

Source: Bernstein Analysis and Estimates

EXHIBIT 16: **Cooling content product coverage by company (1/2)**

✔ Core product ✓ Minor offering ✗ Not offered

Company / Component	VERTIV	nVent	BOYD	Schneider Electric	CoolIT systems	DELTA	AVC	Auras
Chillers	✔	✗	✓	✔	✗	✔	✗	✗
CDU	✔	✔	✔	✔	✔	✔	✓	✓
Cold plates	✓	✗	✔	✓	✔	✓	✔	✔
CRAH	✔	✗	✗	✔	✗	✔	✗	✗
RDHx	✔	✔	✗	✔	✗	✔	✗	✗
Manifolds	✔	✔	✔	✔	✔	✔	✔	✔
Dry coolers	✔	✗	✗	✔	✗	✔	✗	✗

Source: Bernstein Analysis

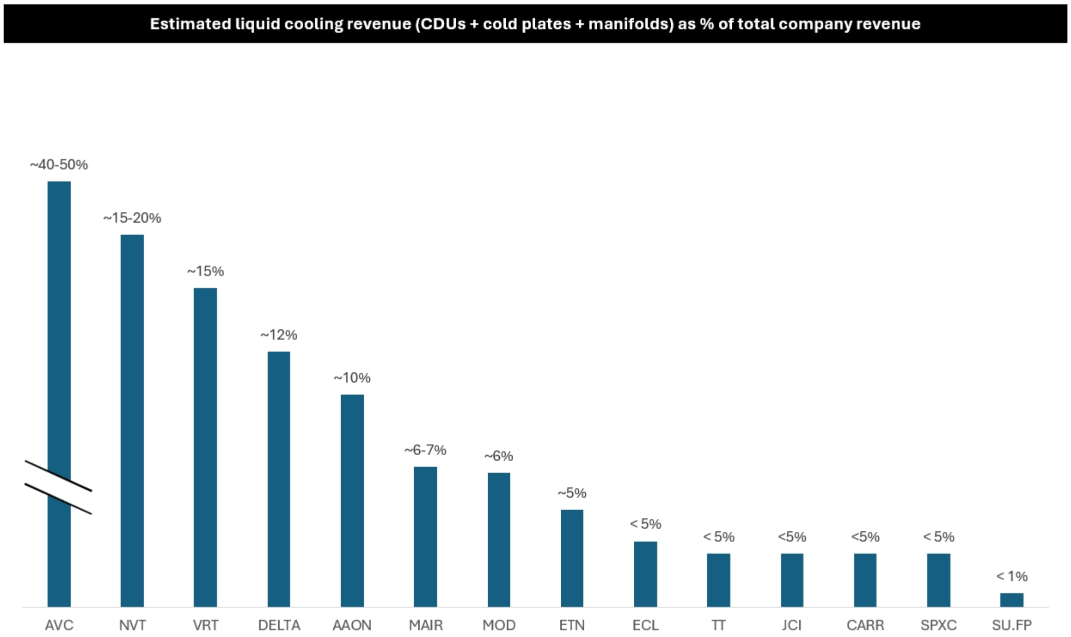
EXHIBIT 17: **Cooling content product coverage by company (2/2)**

✔ Core product ✓ Minor offering ✗ Not offered

Company / Component	TRANE TECHNOLOGIES	Johnson Controls	Carrier	MODINE	MADISONAIR	SPX TECHNOLOGIES	AAON	COOLER MASTER
Chillers	✔	✔	✔	✔	✓	✗	✔	✗
CDU	✔	✔	✔	✔	✔	✗	✔	✔
Cold plates	✗	✗	✗	✗	✗	✗	✗	✔
CRAH	✔	✔	✔	✔	✔	✓	✔	✗
RDHx	✗	✗	✗	✗	✔	✗	✗	✗
Manifolds	✗	✗	✗	✗	✓	✗	✗	✔
Dry coolers	✗	✔	✔	✔	✗	✔	✔	✗

Source: Bernstein Analysis

EXHIBIT 18: **Some companies are more exposed to liquid cooling than others**



Source: Bernstein Analysis and Estimates

RESEARCH INDEX

Primer Library

Liquid Cooling Primer: Coolant Distribution Units (CDUs)

Liquid Cooling Primer: Cold Plates

Data Center Chillers (1/3): Primer and free cooling economics

Data Center Chillers (2/3): How real is Chiller Gate risk?

Data Center Chillers (3/3): How much can "free cooling" impact chiller service revenues?

Data Center Cooling and Power: What's the $M per MW for different types of equipment

Orbital Data Centers: Content ($M/MW) Implications for Cooling Equipment Players

Other Relevant Research

Data Center Cooling Equipment: What is the service tail of a CDU (Coolant Distribution Unit)?

US Multis: Mapping the Data Center Service Footprint for Chiller OEMs (CARR / JCI / TT)

Data Center Cooling: Accelsius on the transition from single-phase to two-phase liquid cooling

Liquid Cooling: What does Google's Brazos mean for the broader CDU ecosystem?

Liquid Cooling in Data Centers: What comes after single phase DTC? With CoolIT's 15kW cold plate, does anything even need to?

Data Center Cooling: Chiller killer? Quick thoughts on Accelsius' two-phase cooling at >54 degree celsius inlet temperatures.

AI Data Centers: Procurement dynamics across power and cooling components (webinar recap)

US Industrials: 30 exhibits on the state of cooling and electrical equip. procurement in North American Data Centers

Initiations

US Multis Initiation: Thrills and Chills

US Multi Industrials: Can we (inter)connect? Initiating coverage with Outperform on Amphenol and TE Connectivity

The authors would like to thank Sakansh Mittal for his help in preparing this report.

I. REQUIRED DISCLOSURES

References to "Bernstein" or the “Firm” in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK（サンフォード・C・バーンスタイン株式会社）and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein’s “affiliates” relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY

This research publication covers six or more companies. For valuation methodology and other company disclosures:
Please visit: https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.
Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RISKS

This research publication covers six or more companies. For risks and other company disclosures:
Please visit: https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.
Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/-15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as ‘Feature’ (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	52.3%	14.2%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.6%	16.1%
Underperform	SELL	12.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of September 30, 2026. All figures are updated quarterly.

PRICE CHARTS/ RATINGS AND PRICE TARGET HISTORY

This research publication covers six or more companies. For price chart and other company disclosures, please visit https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action or you can write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

CONFLICTS OF INTEREST

SG and/or an affiliate is acting in the expected equity issuance of Schneider Electric via an Accelerated Bookbuild Offering (ABO) to refinance partially the bridge facility of its acquisition of PTC Inc.
Bernstein Autonomous LLP or BSG France SA, beneficially, has either a net long or short position of 0.5% or more of the total issued share capital of a class of common equity securities of the following MiFID eligible securities: Trane Technologies PLC, Carrier Global Corporation, Eaton Corp PLC and Digital Realty Trust, Inc..

AB and/or its affiliates beneficially own 1% or more of a class of common equity securities of the following companies: Eaton Corp PLC and Blackstone Digital Infrastructure Trust Inc..

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Blackstone Digital Infrastructure Trust Inc., Csquare, Inc. and Schneider Electric SA.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Carrier Global Corporation, Equinix, Inc. and Schneider Electric SA.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Equinix, Inc..

Bernstein and/or affiliates expect to receive or intend to seek compensation for investment banking services in the next three months from Csquare, Inc. and Schneider Electric SA.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Vertiv Holdings Co, nVent Electric PLC, Trane Technologies PLC, Johnson Controls International PLC, Carrier Global Corporation, Eaton Corp PLC and Schneider Electric SA.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: Carrier Global Corporation, Eaton Corp PLC, Digital Realty Trust, Inc., Equinix, Inc. and Schneider Electric SA.

Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Blackstone Digital Infrastructure Trust Inc., Csquare, Inc. and Schneider Electric SA.

Affiliates of Bernstein managed or co-managed in the past twelve months a public offering of securities of Blackstone Digital Infrastructure Trust Inc., Csquare, Inc. and Schneider Electric SA.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

+1 Bernstein Institutional Services LLC; New York, New York, USA

+44 Bernstein Autonomous LLP; London UK

+212 Société Générale Global Solution Center Morocco, Casablanca, Morocco

+33 BSG France S.A.; Paris, France

+34 BSG France S.A.; Madrid, Spain

+41 Bernstein Autonomous LLP; Geneva, Switzerland

+49 BSG France S.A.; Frankfurt, Germany

+91 Sanford C. Bernstein (India) Private Limited; Mumbai, India

+852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China

+65 Sanford C. Bernstein (Singapore) Private Limited; Singapore

+81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for “Bernstein” and “Autonomous” research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the “Autonomous” and “Bernstein” brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.

- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to “Autonomous” branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation (“MAR”) and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission (“SEC”) and a member of the U.S. Financial Industry Regulatory Authority, Inc. (“FINRA”) is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the “Financial Promotion Order”), (ii) are persons falling within Article 49(2)(a) to (d) (“high net worth companies, unincorporated associations, etc.”) of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as “relevant persons”). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (https://www.sfc.hk/publicregWeb/corp/AXC846/details)). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 202137710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People’s Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the “PRC”) in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. (“Daiwa”), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;

- dealing in a financial product;

- making a market for a financial product; and

- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).

- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com / scbincompliance@bernsteinsg.com.

- SCB India does not have any disciplinary history as on the date of this report.

- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives

  - do not have any financial interest in the subject company

  - do not have actual/beneficial ownership of one percent or more in securities of the subject company;

  - is not engaged in any investment banking activities for Indian companies, as such;

  - have not managed or co-managed a public offering in the past twelve months for any Indian companies;

  - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;

  - have not received compensation for brokerage services from the subject company in the past twelve months;

  - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and

  - do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.

  - do not have any conflict of interest in the subject company as of the date of this report.

- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.

- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.

- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com / scbincompliance@bernsteinsg.com.

- Kindly refer to the Research investor charter and Research Most Important Terms & Conditions (MITC) available at (https://bernsteinresearch.in/)

- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.